

EX.NO: 03

Exploring the Different Types of Prompts

Straightforward Prompt · Tabular Format Prompting · Preceding Question Prompting · Missing Word Prompting

1. Aim & Objective

The aim of this experiment is to understand and explore different prompt engineering techniques used while interacting with Artificial Intelligence systems. Prompt engineering helps users obtain accurate, structured, and meaningful responses from AI models by designing effective prompts.

In this experiment, four important prompt types are studied:

- 1 Straightforward Prompts
- 2 Tabular Format Prompting
- 3 Preceding Question Prompting
- 4 Missing Word Prompting

For each prompt type, different prompts are created using multiple real-world scenarios. These prompts demonstrate how the structure of a question affects the quality, clarity, and depth of the AI-generated response.

2. Introduction

Prompting is the process of giving instructions or questions to an AI model in order to generate the required response. Different prompting techniques are useful for different situations. Some prompts are best for short answers, while others are

better for explanations, comparisons, or educational exercises.

Prompt engineering plays an important role in:

- Artificial Intelligence applications
- Healthcare and medical systems
- Education and e-learning platforms
- Data analysis and research
- Content generation and summarisation
- Chatbot and virtual assistant interactions

By using proper prompts, users can improve response accuracy, readability, and contextual understanding.

3. Prompt Type 1 – Straightforward Prompts

Explanation

Straightforward prompts are direct and simple instructions given to the AI model. These prompts clearly ask for a specific answer without additional context or examples. They are mainly used when the user wants quick, factual, and concise information.

Straightforward prompting is useful for:

- Definitions
- Quick explanations
- Listing features or advantages
- Simple factual responses

3.1 Use Case A – AI in Healthcare

Scenario:

A medical student wants quick explanations of important Artificial Intelligence concepts used in the healthcare domain.

Prompt 1:**Prompt:**

"Define Artificial Intelligence in Healthcare in one sentence."

AI Response:

Artificial Intelligence in Healthcare refers to the use of machine learning algorithms and data analytics to assist in medical diagnosis, treatment planning, and patient care.

Prompt 2:**Prompt:**

"List three applications of AI in the medical field."

AI Response:

- 1 Medical image analysis for detecting tumours in X-rays and MRI scans
- 2 Predictive analytics for identifying high-risk patients
- 3 AI-powered virtual health assistants for patient support

Prompt 3:**Prompt:**

"What is an Electronic Health Record (EHR)?"

AI Response:

An Electronic Health Record (EHR) is a digital version of a patient's medical history maintained over time by healthcare providers, containing diagnoses, medications, and treatment records.

3.2 Use Case B – AI in Education

Scenario:

A teacher wants quick educational definitions of AI-related concepts for preparing classroom notes and study materials.

Prompt 1:

Prompt:

"Define adaptive learning in one sentence."

AI Response:

Adaptive learning is an AI-driven educational approach that adjusts the difficulty, pace, and style of instruction based on individual student performance and learning needs.

Prompt 2:

Prompt:

"List four AI tools commonly used in education."

AI Response:

- 1 Duolingo – AI-powered language learning
- 2 Knewton Alta – adaptive course content delivery
- 3 Grammarly – AI writing assistant
- 4 Socratic by Google – step-by-step homework help

Prompt 3:

Prompt:

"What is the role of Natural Language Processing in education?"

AI Response:

Natural Language Processing (NLP) in education enables AI systems to understand and evaluate student-written responses, provide grammar feedback, and power intelligent tutoring chatbots that answer students' questions in real time.

4. Prompt Type 2 – Tabular Format Prompting

Explanation

Tabular format prompting instructs the AI to organise information into rows and columns. This method improves readability and makes comparisons easier.

Tabular prompts are useful for:

- Comparing multiple items
- Organising structured data
- Presenting features clearly
- Study references and reports

4.1 Use Case A – AI in Healthcare

Scenario:

A medical researcher wants to compare AI-assisted diagnosis with traditional diagnosis methods to understand the benefits of modern technology.

Prompt:

“Create a table comparing Traditional Diagnosis and AI-Assisted Diagnosis with columns for Aspect, Traditional Diagnosis, and AI-Assisted Diagnosis.”

Aspect	Traditional Diagnosis	AI-Assisted Diagnosis
Speed	Depends on physician availability	Real-time analysis

Accuracy	Prone to human error	High accuracy via pattern recognition
Cost	High consultation fees	Cost-effective at scale
Data Handling	Manual record keeping	Automated EHR integration
Availability	Limited to clinic hours	Available 24/7

Observation:

The tabular format makes it easy to compare key differences between traditional and AI-assisted diagnosis methods in healthcare.

4.2 Use Case B – AI in Education

Scenario:

An education researcher wants to compare popular AI-powered learning tools to understand their features and benefits.

Prompt:

“Provide a table listing four AI tools used in education with columns for Tool Name, Type of AI Used, and Primary Benefit.”

Tool Name	Type of AI Used	Primary Benefit
Duolingo	NLP + Adaptive ML	Personalised language learning
Knewton Alta	Adaptive Learning Engine	Customised course pacing
Grammarly	NLP + Deep Learning	Real-time writing feedback
Socratic by Google	Computer Vision + NLP	Step-by-step homework help

Observation:

The table provides a clear comparison of AI tools used in education, highlighting the specific technology and benefit of each tool.

4.3 Use Case C – Traditional vs AI-Powered Learning

Prompt:

“Create a table comparing Traditional Classroom Learning and AI-Powered E-Learning with columns for Feature, Traditional Learning, and AI-Powered E-Learning.”

Feature	Traditional Learning	AI-Powered E-Learning
Personalisation	One-size-fits-all	Tailored to each learner
Feedback	Delayed (exams)	Instant and continuous
Accessibility	Fixed location & time	Anytime, anywhere
Assessment	Periodic standardised tests	Continuous adaptive assessment

Observation:

The table provides a clear comparison and improves understanding of the advantages of AI-powered e-learning over traditional classroom learning.

5. Prompt Type 3 – Preceding Question Prompting

Explanation

Preceding Question Prompting uses a two-step questioning approach. First, a general question is asked to activate context. Then a second question asks for a detailed explanation.

This prompting style is useful for:

- Detailed explanations
- Conceptual understanding
- Analytical answers

- Academic discussions

5.1 Use Case A – AI in Healthcare

Scenario:

A medical student wants a detailed understanding of how AI helps with early disease detection.

Prompt:

“Why is early disease detection critical in healthcare? Explain how AI-powered diagnostic tools help doctors identify diseases at an early stage.”

AI Response:

Early disease detection is critical because it significantly increases the chances of successful treatment, reduces healthcare costs, and prevents complications. AI-powered diagnostic tools analyse medical images, lab results, and patient history using deep learning models. These tools can identify abnormal patterns that may be invisible to the human eye, enabling doctors to detect cancers, heart conditions, and other diseases at an early and treatable stage.

5.2 Use Case B – AI in Education

Scenario:

A researcher wants to understand the problems with traditional learning and how AI-based adaptive systems can solve them.

Prompt:

“Why do students struggle with one-size-fits-all teaching methods? Explain how AI-driven adaptive learning platforms can address these challenges and improve student outcomes.”

AI Response:

Students struggle with uniform teaching methods because every learner has different strengths, weaknesses, and learning speeds. Traditional classrooms cannot accommodate these individual differences. AI-driven adaptive learning platforms solve this by continuously analysing student performance data and adjusting the content, difficulty, and pace of lessons accordingly. This personalised approach keeps students engaged, reduces learning gaps, and improves overall academic outcomes.

5.3 Use Case C – AI and Data Privacy in Healthcare

Prompt:

“Why is patient data privacy important in AI-driven healthcare systems? Explain how encryption and secure data practices protect sensitive medical information.”

AI Response:

Patient data privacy is important because medical records contain sensitive personal information that, if breached, can lead to identity theft, discrimination, or misuse. In AI-driven healthcare systems, large volumes of data are collected and processed, making security critical. Encryption converts data into unreadable code that can only be accessed with a key, while secure data practices such as access control, audit logs, and anonymisation ensure that patient information is used only for its intended purpose.

6. Prompt Type 4 – Missing Word Prompting

Explanation

Missing Word Prompting uses fill-in-the-blank style questions where the AI must complete missing terms. These prompts are commonly used in quizzes, educational tests, and vocabulary exercises.

This method helps in:

- Testing knowledge

- Educational assessments
- Memory recall
- Practice exercises

6.1 Use Case A – AI in Healthcare

Prompt:

"Complete the following sentences:

- 1 In AI-assisted radiology, deep learning models are trained to detect ____ in medical images.
- 2 The process of using patient data to forecast future health risks is called ____ analytics.
- 3 An AI system that reads doctors' clinical notes and converts them into structured records uses ____.

AI Response:

- 1 Anomalies / tumours
- 2 Predictive
- 3 Natural Language Processing (NLP)

6.2 Use Case B – AI in Education

Prompt:

"Fill in the blanks:

- 1 An AI learning platform that adjusts difficulty based on student performance is called a(n) ____ learning system.
- 2 Grammarly uses ____ language processing to provide real-time grammar suggestions.
- 3 AI tools like Socratic by Google use ____ vision to read and solve handwritten problems.

AI Response:

- 1 Adaptive
- 2 Natural
- 3 Computer

6.3 Use Case C – General AI Concepts

Prompt:

"Complete the following:

- 1 The branch of AI that allows machines to learn from data without explicit programming is called ____.
- 2 A neural network inspired by the structure of the human brain is called a ____ neural network.
- 3 The full form of NLP is ____.

AI Response:

- 1 Machine Learning
 - 2 Deep / Artificial
 - 3 Natural Language Processing
-

7. Summary & Observations

Prompt Type	Main Purpose	Key Observation
Straightforward Prompt	Direct answers	Gives concise responses quickly
Tabular Format Prompting	Organising data	Improves readability and comparison
Preceding Question Prompting	Detailed explanation	Produces context-rich responses
Missing Word Prompting	Knowledge testing	Useful for quizzes and educational exercises

8. Conclusion

This experiment successfully demonstrated the use of different prompt engineering techniques. Each prompt type serves a unique purpose depending on the user's requirement.

- Straightforward prompts are best for direct and factual information.
- Tabular prompts help organise and compare data effectively.
- Preceding question prompts provide detailed conceptual explanations.
- Missing word prompts are ideal for educational assessments and quizzes.

Therefore, understanding different prompt types helps users communicate more effectively with AI systems and obtain better-quality responses.